

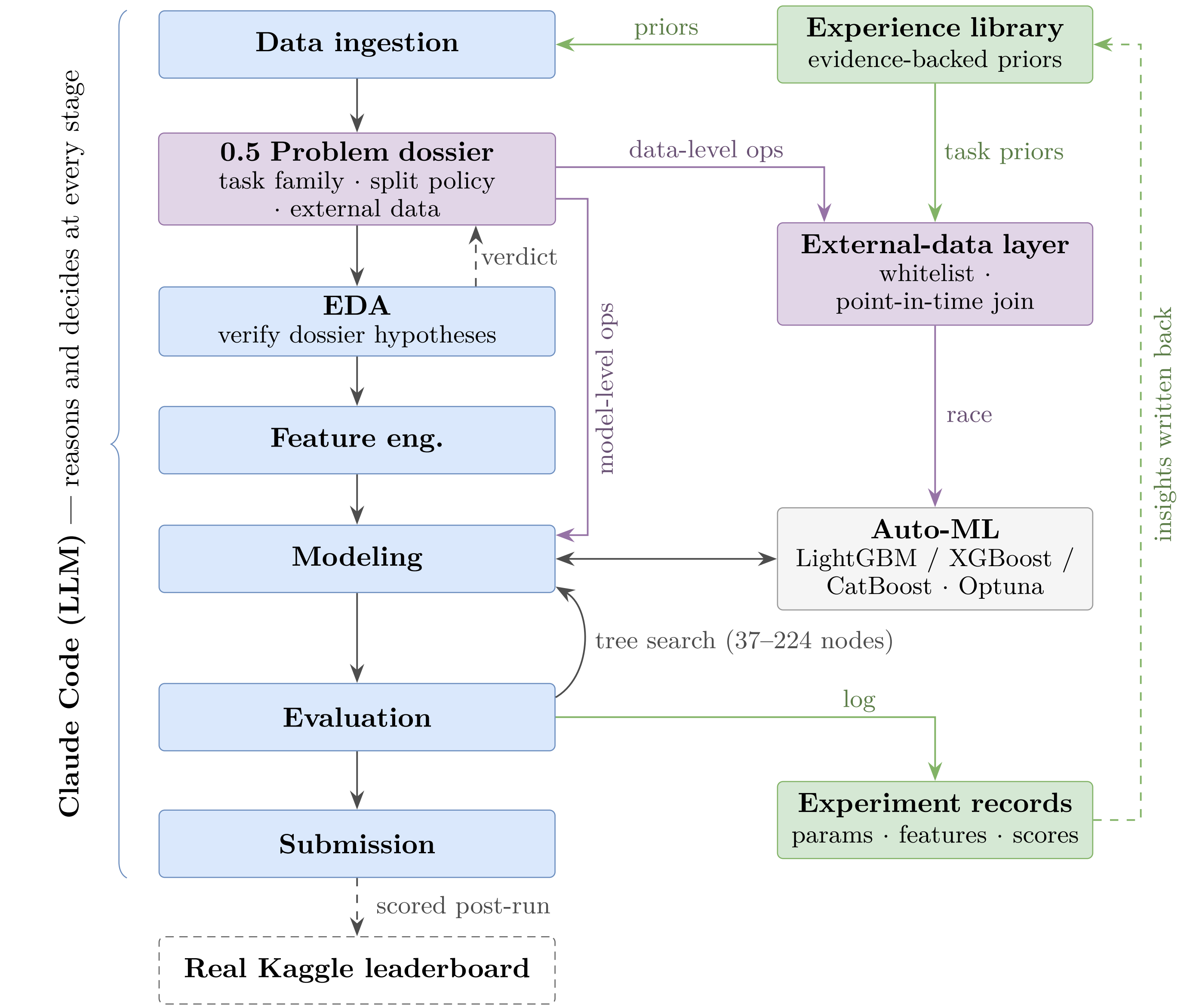
An End-to-End AI Agent for Kaggle Competitions

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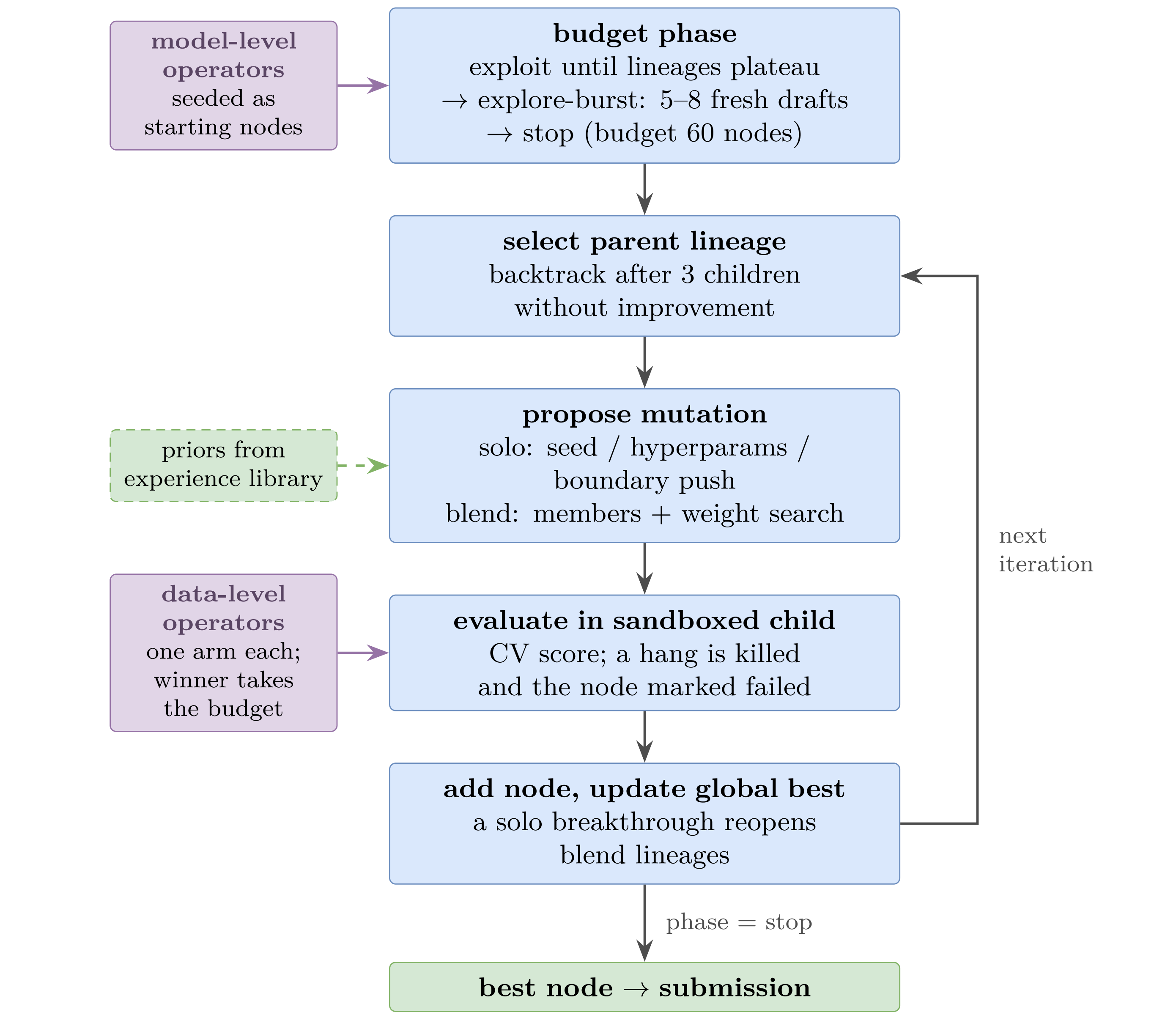
1 · Motivation

- Challenges of Kaggle competitions
 - A competition requires the expert loop**
This includes validation design, feature extraction, model selection, and ensemble building. The top entries separate by a very small margin.
 - Data preparation is repetitive**
The same cleaning, split design and drift are done on every dataset.
- Why an AI agent
 - A competition is a *scorable* task**
Every competitor delivers a number, so it can be optimized automatically.
 - A competition requires tireless, systematic search**
Every experiment is recorded and should be reproducible.
 - Recent success in agents for Kaggle competitions**
Notable cases include tree search over code [1], and tree search fed injected research ideas [2].

2 · Workflow



The LLM reasons and decides at every stage; the **problem dossier** classifies the task and injects external data *upstream*, before any modeling. EDA then tests the dossier's hypotheses, feature engineering and modeling follow, and evaluation returns the score the tree search uses to pick the next candidate. Purple = injection layer · green = evidence loop · gray = Auto-ML tools; dashed = feedback or post-run.



3 · Results

Our agent “Sinica” were carried out in audited sandboxes for all 23 competitions (20 ordinary competitions in Table 1 and 3 special competitions in Table 2). The results of the 20 ordinary competitions were scored on the real leaderboard, while the results of the 3 special competitions were scored offline with MLE-bench [4]. We compared these results with the results delivered by two other agents AIDE [1] (tree search over code) and the NVIDIA Kaggle Agent [3] (kernel reproduction). **Among the 23 competitions, our agent “Sinica” obtains the best performance with 13 wins and 10 losses.**

Remark: MLE-bench is an offline benchmark that replays finished Kaggle competitions with their original data and metrics.

Table 1: ordinary competitions

Competition	Metric	Sinica	NVIDIA	AIDE
s3e1	RMSE ↓	0.55275	0.55550	0.55552
s3e3	AUC ↑	0.87157	0.89537	0.86863
s3e5	QWK ↑	0.57223	0.56974	0.58138
s3e7	AUC ↑	0.90441	0.90458	0.90109
s3e9	RMSE ↓	12.3319	12.2291	12.3067
s3e11	RMSLE ↓	0.29368	0.29624	0.29445
s3e14	MAE ↓	333.186	330.716	332.316
s3e16	MAE ↓	1.33822	1.34001	1.34224
s3e19	SMAPE ↓	49.4088	48.2656	48.3413
s4e11	Acc. ↑	0.94107	0.94076	0.94056
s5e1	MAPE ↓	0.08401	0.14594	0.17917
s5e10	RMSE ↓	0.05576	0.05588	0.05579
s6e1	Error ↓	8.71516	8.73239	8.72191
s6e2	AUC ↑	0.95505	0.95508	0.95516
afsis	MCRMSE ↓	0.48917	0.69855	0.49861
cat-in-the-dat	AUC ↑	0.80243	0.77084	0.80217
conway	MAE ↓	0.10414	0.11189	0.11065
tps-aug-2022	AUC ↑	0.58930	0.58730	0.59106
tps-jan-2022	SMAPE ↓	6.29895	4.63651	10.6664
tps-sep-2022	SMAPE ↓	28.4961	5.48351	9.72095
Best private score		10 / 20	7 / 20	3 / 20

Bold = best of the three.

Table 2: special competitions

Competition	Metric	Sinica	NVIDIA	AIDE
us-patent	Pearson r ↑	0.86658	0.86160	0.65821
ventilator	MAE ↓	0.16525	0.40093	0.34591
SIIM-ISIC	AUC ↑	0.93529	0.89822	0.88579

Graded by MLE-bench.

4 · Disadvantages of the agent-based approach

- Strong at coding, weak at judging**
The agent efficiently works through whatever problem it is given. However, it does not ask whether a change is worth making. Therefore, a poor design choice shows up and fixing the bugs caused by it is costly.
- The main question gets buried in detail**
AI-written reports pile up technical detail and drift away from what the reader wants to know.
- AI writing habits are difficult to remove**
Habits such as redundant headings and invented terms come back in every draft; verifying and modifying them by hand is exhausting.

5 · Practical advice for building AI agents for data science

- Justify the design before building**
Settle on a solid design before the runs start; a redesign costs not only debugging but also a full re-run of every earlier competition.
- Put external knowledge before modeling**
Injecting external data at the final blending step changed scores by almost nothing. Therefore, we moved the injection to before modeling, and scores improved clearly.
- Build isolation into the environment**
Set up the sandbox to prevent any access to the leaderboard or another run's files.
- Iteration can only target the local validation score**
The local validation score is the only target the agent can iterate on. As a result, the local validation score can look outstanding while the real leaderboard score is mediocre.

6 · References

- Z. Jiang, D. Schmidt, D. Srikanth, D. Xu, I. Kaplan, D. Jacenko, and Y. Wu, AIDE: AI-driven exploration in the space of code, *arXiv:2502.13138*.
- E. Aygün *et al.*, An AI system to help scientists write expert-level empirical software, *Nature* **654**, 909 (2026).
- NVIDIA Corporation, computer code NVIDIA Kaggle Plugin, Santa Clara, CA, 2026, <https://github.com/NVIDIA/nvidia-kaggle>.
- J. S. Chan *et al.*, MLE-bench: Evaluating machine learning agents on machine learning engineering, in *Proceedings of the 13th International Conference on Learning Representations*, 2025: <https://openreview.net/forum?id=6s5uXNWGIh>.